Predicting self-regulated learning support needs

Rick Dijkstra, dr. Max Hinne, prof. dr. Eliane Segers, Josh Ring, prof. dr. Inge Molenaar

Radboud University

Introduction
Self-regulated learning support

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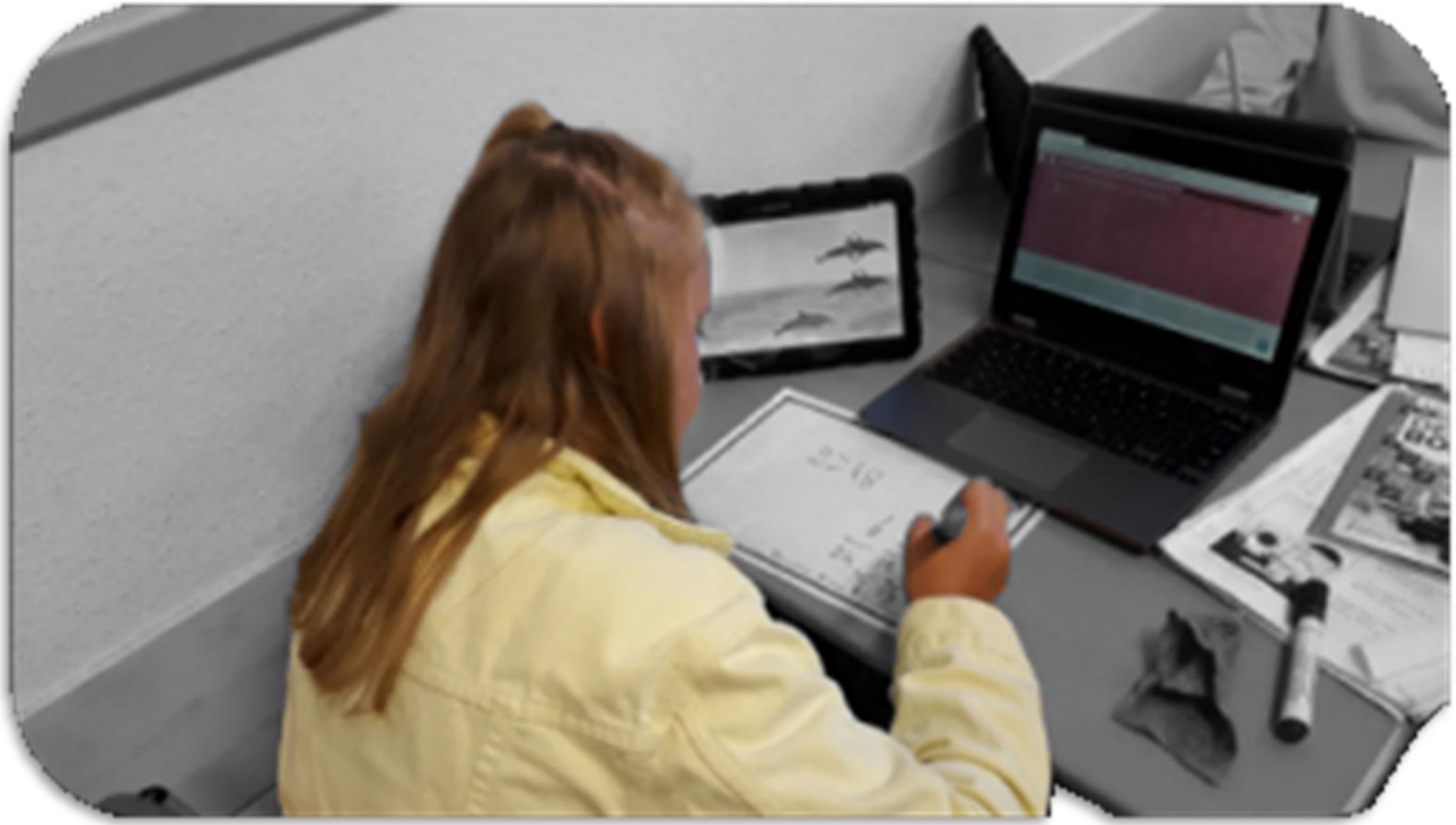
Self-regulated learning support



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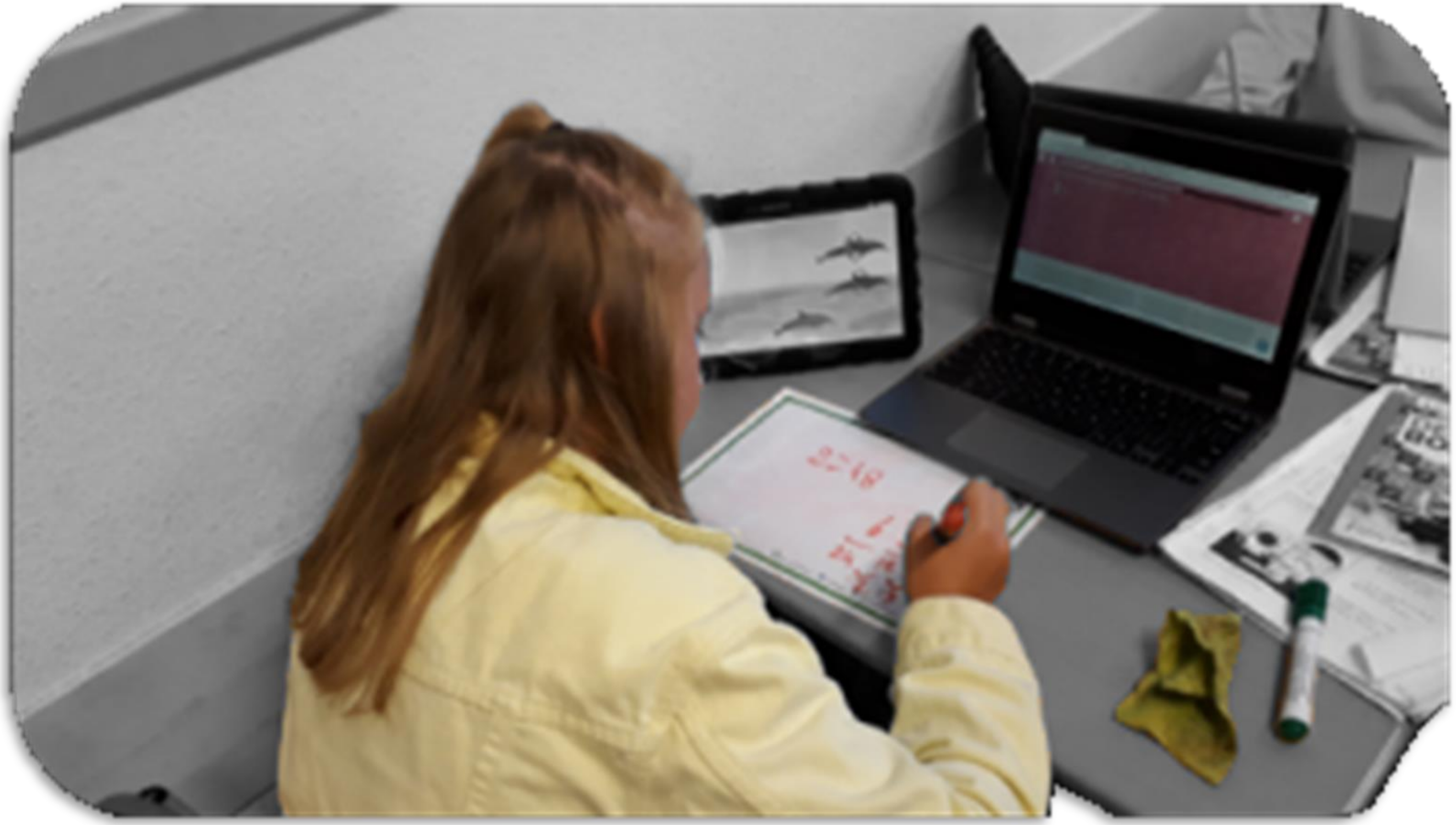
Self-regulated learning support



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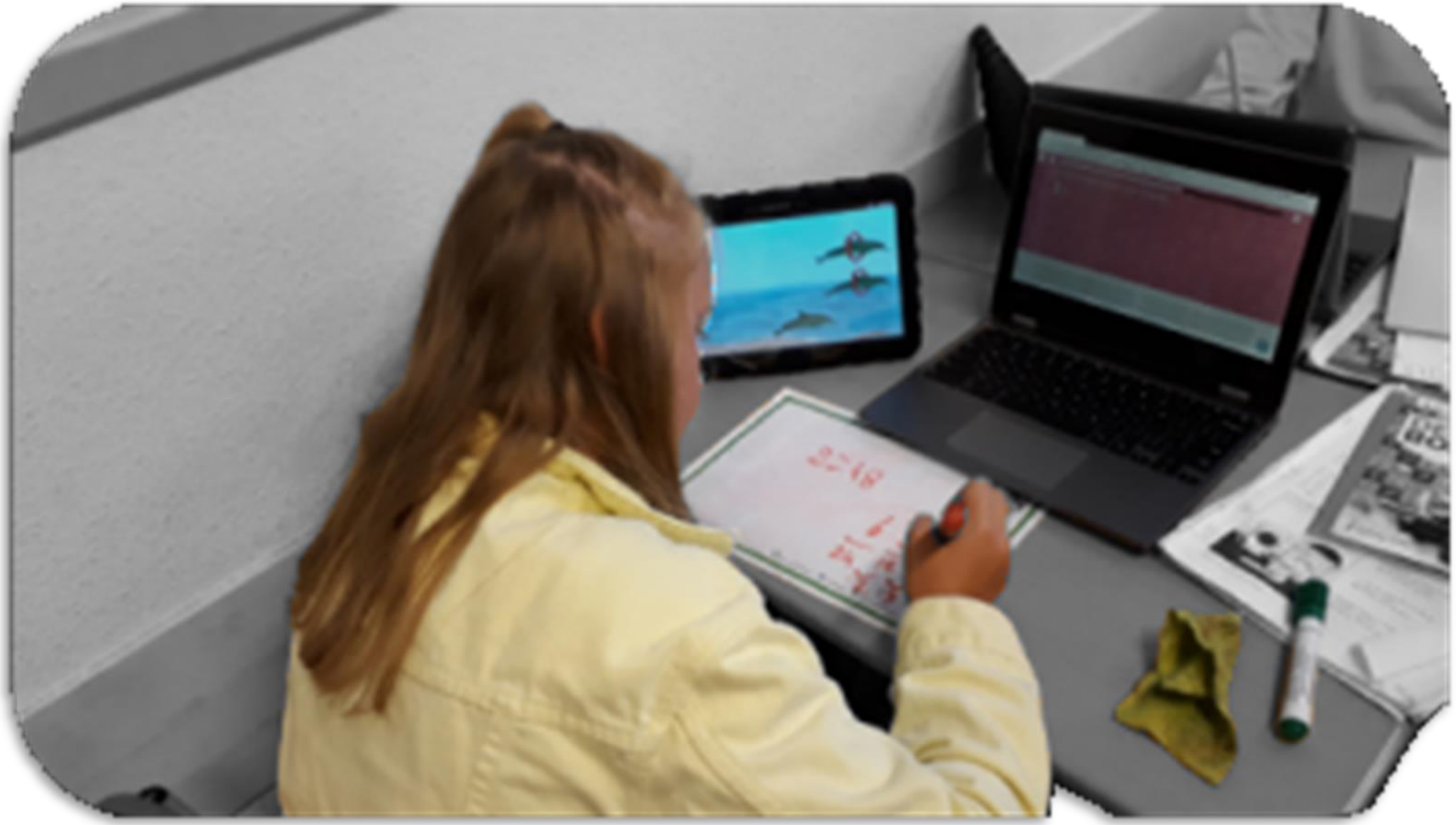
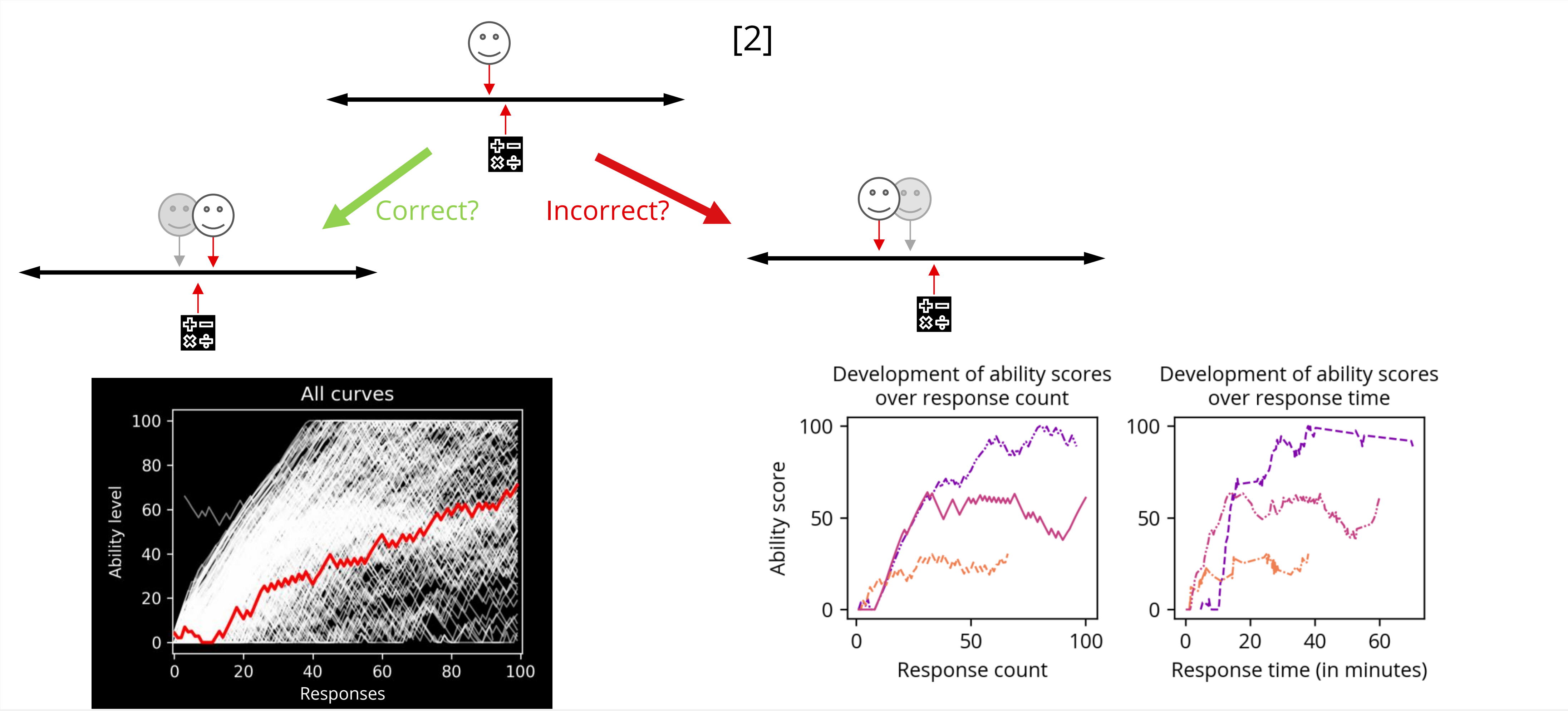
Self-regulated learning support



AIMS:

1) DETECT SRL SUPPORT NEEDS
SO THAT WE CAN
2) PREDICT SRL SUPPORT NEEDS
DURING LEARNING

Methods
Log data



Measurements

Pre test

Number of correct answers

4 days with 30 minutes of practice

First 3 days one learning goal each, 4th day repetition

Accuracy (= Percentage of correct responses)
Effort (= Total number of responses)
Learning curve

Post test

Number of correct answers

Methods
1) Detecting SRL support needs: Clustering [1]

Bayesian non-parametric clustering

Learn the amount of clusters from the data
Learn the shape of the clusters from the data
Cluster based on the whole ability curve

Identify support needs for clusters

Based on learning outcomes and process measures

Methods

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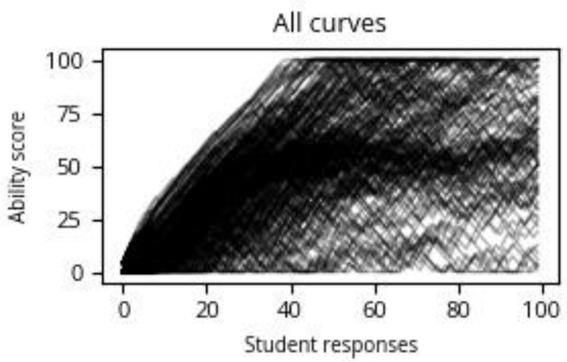

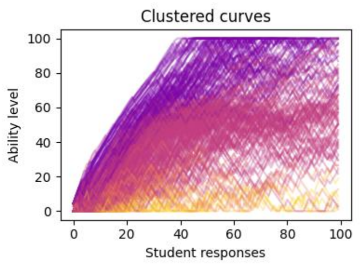

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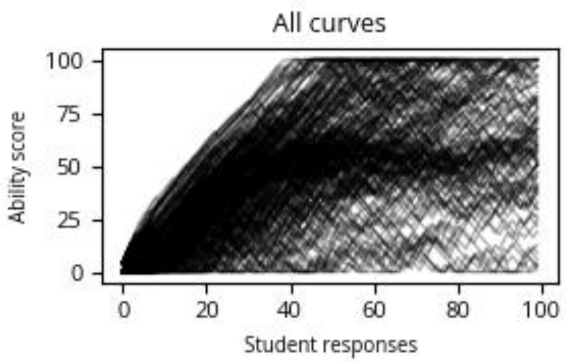

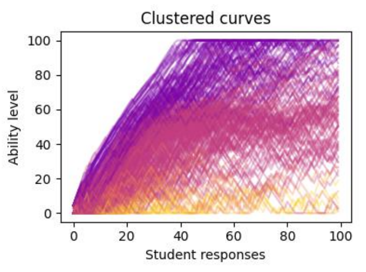

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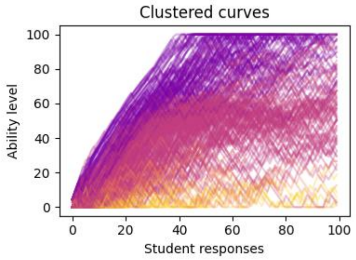

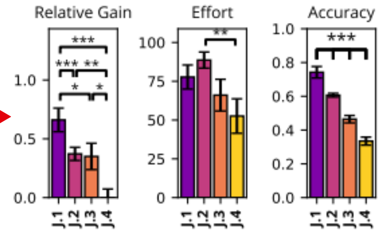

Support needs 1
Support needs 2
Support needs ...
Support needs N

Methods
2) Predicting SRL support needs

Assign to cluster based on partial curve

Partial curve: the data available at a certain point in learning

Correct prediction if partial curve classification is the same as the full curve classification

Accuracy as Matthews Correlation Coefficient (MCC)

Balanced for different class sizes.

Range from -1 to 1; Aim for 0.75

$$\boxed{\mathrm{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}}$$

RESULTS

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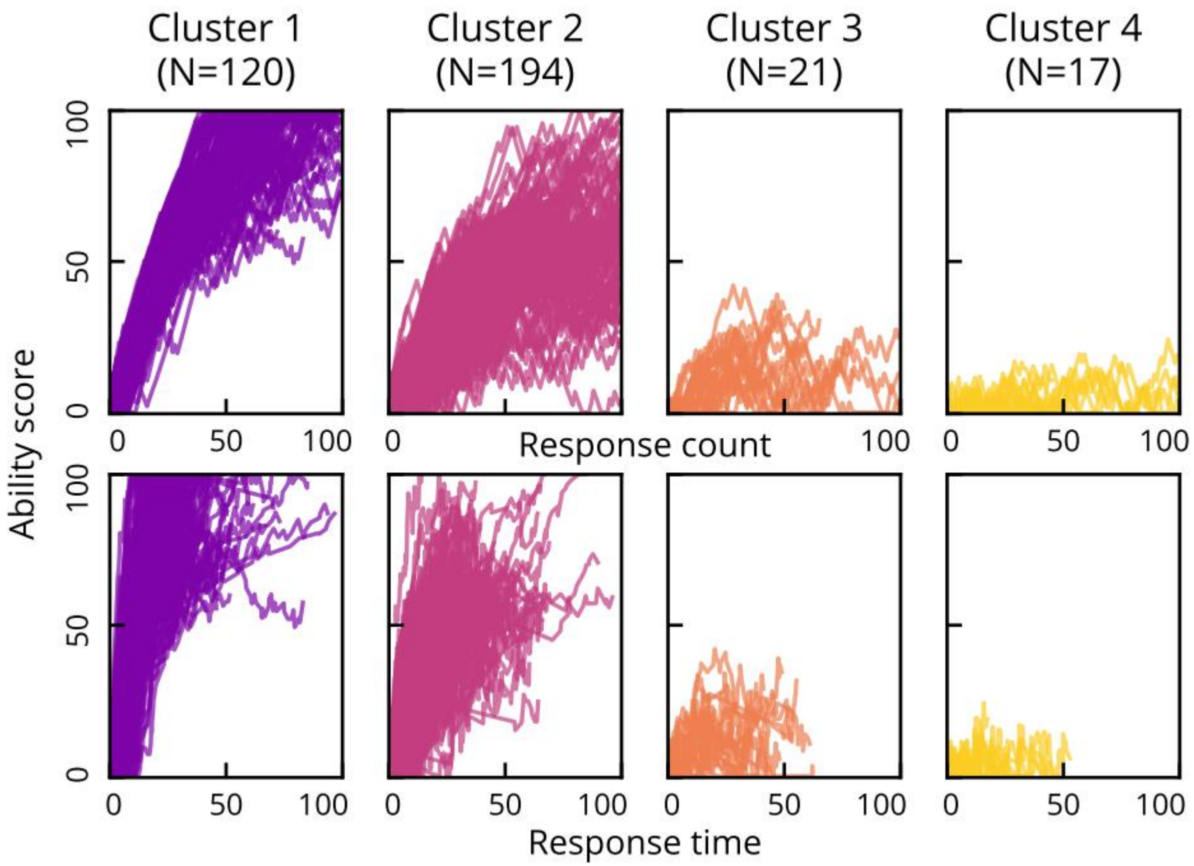
1) DETECTING

1) Detecting - Clusters

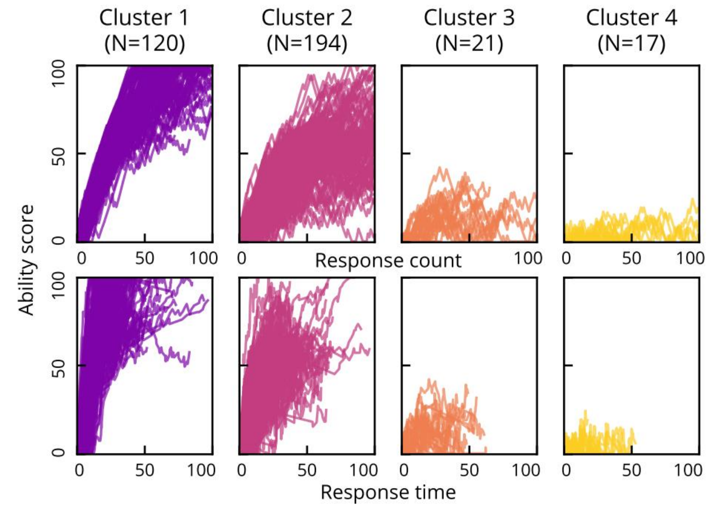
Four clusters

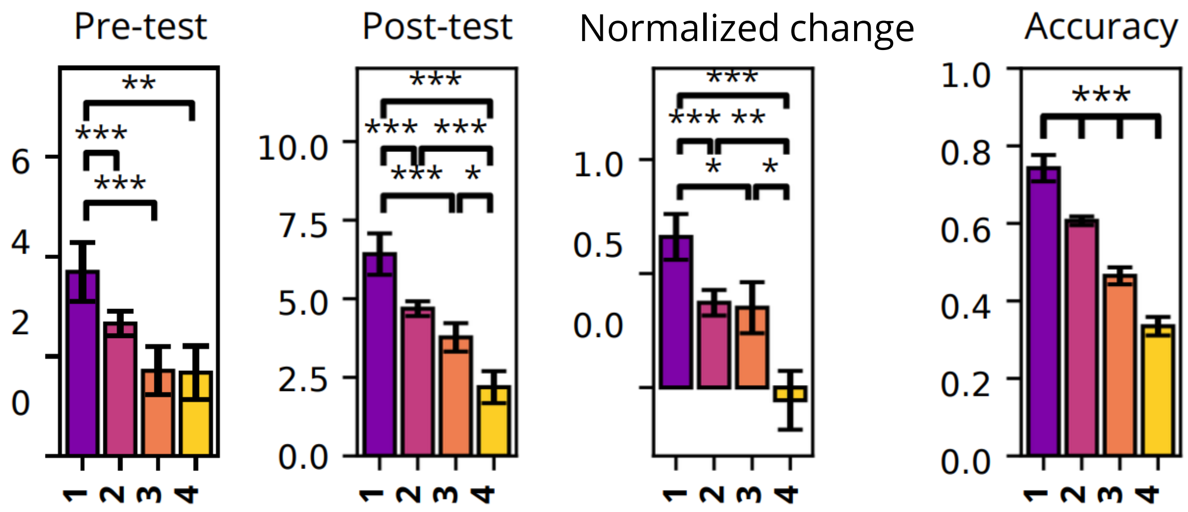
- Plotted for both input dimensions
- Different sizes
- Cluster 3 and 4 looking similar



Results

1) Detecting - SRL support needs





$$\text{Normalized change} = \begin{cases} \dfrac{\text{post} - \text{pre}}{\text{pre}} & \textit{if}\ \text{post} < \text{pre} \\ \dfrac{\text{post} - \text{pre}}{\text{pre}_{max} - \text{pre}} & \textit{if}\ \text{post} > \text{pre} \end{cases}$$

1: Masters
(Minimal support needs)

2: Risers
(Moderate support needs)

3: Strugglers
(Extensive support needs)

4 Trailers
(Extensive support needs)

RESULTS

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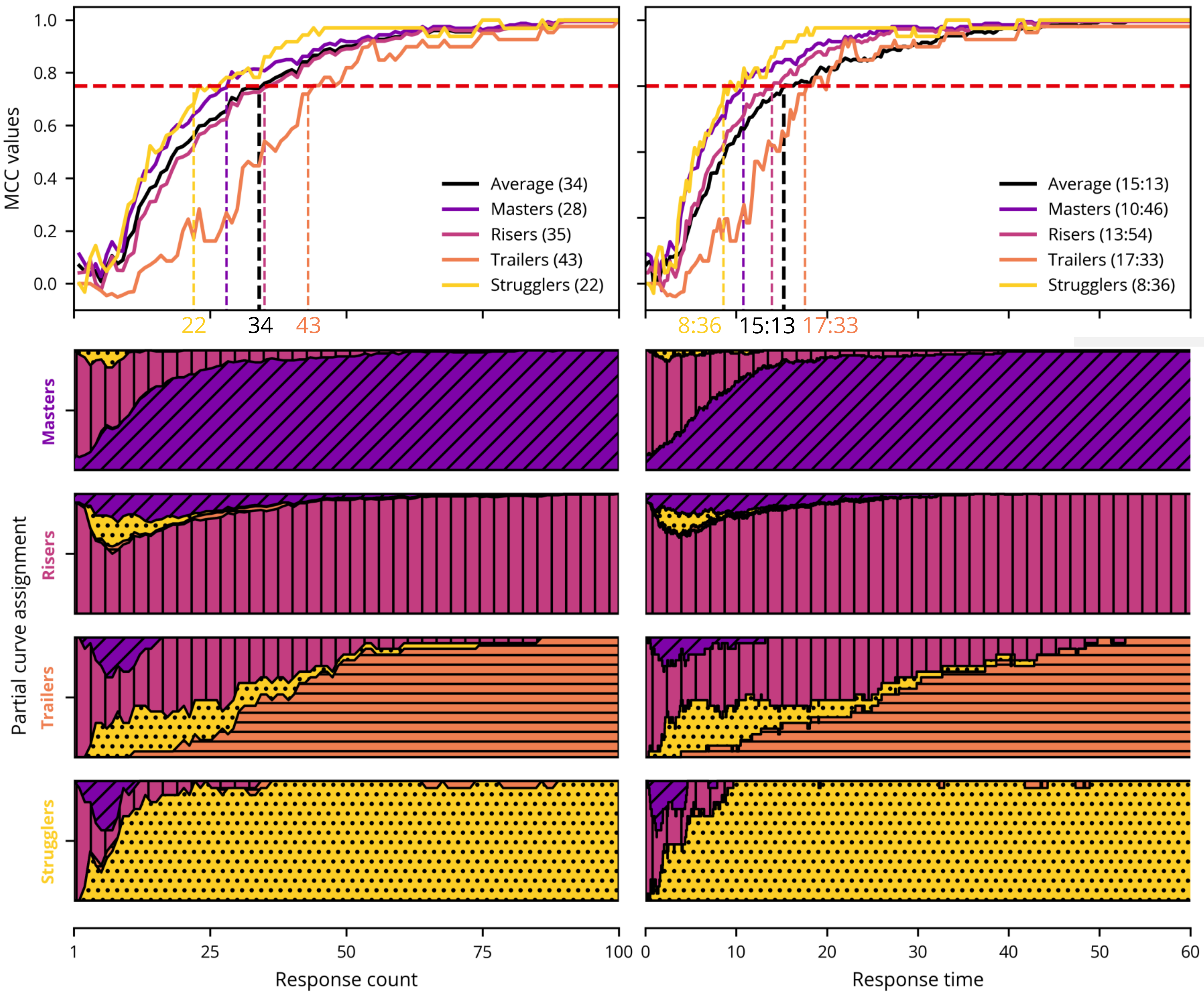
2) PREDICTING

Results

2) Predicting - Accuracy

Average responses in a lesson: 71

One lesson: 30 minutes

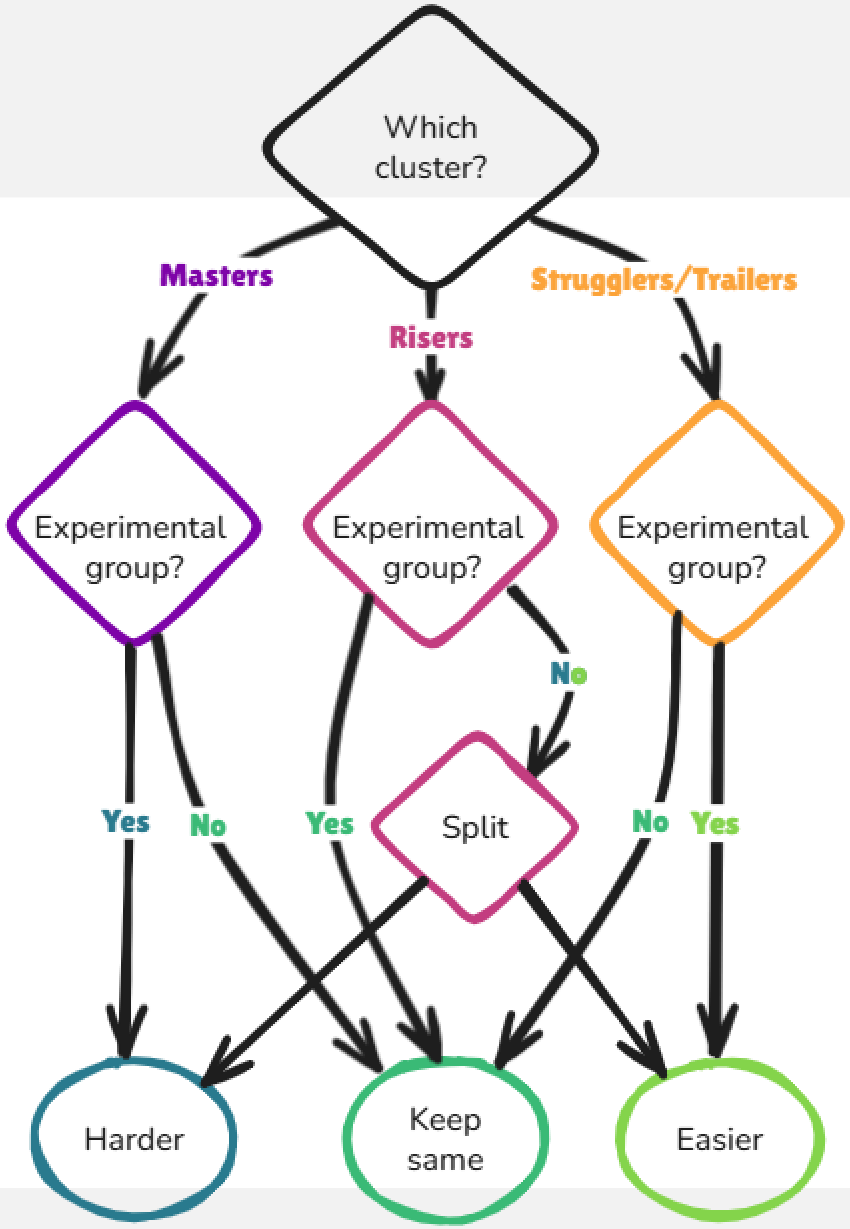


Discussion

Time input is messy, since classrooms are messy.

Prediction makes intervention after 15 minutes possible

Next step: intervention study



THANK YOU!

References

1. Dijkstra R., Hinne M., Segers, E., Molenaar I.: Clustering children's learning behaviour to identify self-regulated learning support needs. Computers in Human Behaviour p.107754 (2023) https://doi.org/10.1016/j.chb.2023.107754
2. Klinkenberg. S., Straatemeijer, M., van der Maas, H.L.J.: Computer adaptive practice of Maths ability using a new item response model for on the fly ability and difficulty estimation. Computers & Education **57**, 1813-1824 (2011). https://doi.org/10.1016/j.compedu.2011.02.003